The Psychology of E-Learning

Individual Characteristics and Web-based Courses
(from the chapter by Wiley & Schooler)

Aptitude-treatment interaction (ATI)

- Aptitude-treatment interaction
  - ATI focuses on performance and how it is influenced by the individual characteristics of the student when the presentation format differs.
  - ATI suggests that optimal learning occurs when the instruction is exactly matched to the aptitudes of the learner.

Presentation Medium

- Most of the comparative studies between instructional media---face to face teaching; paper, television, video, or computer-based formats; and various distance delivery media— **have found no significant difference in student learning.**

- **The questions that now need to be asked are not which medium works best, but rather, how best to incorporate media attributes and individual characteristics into the design of effective instruction.**

Interactive model of learner characteristics and web-based courses

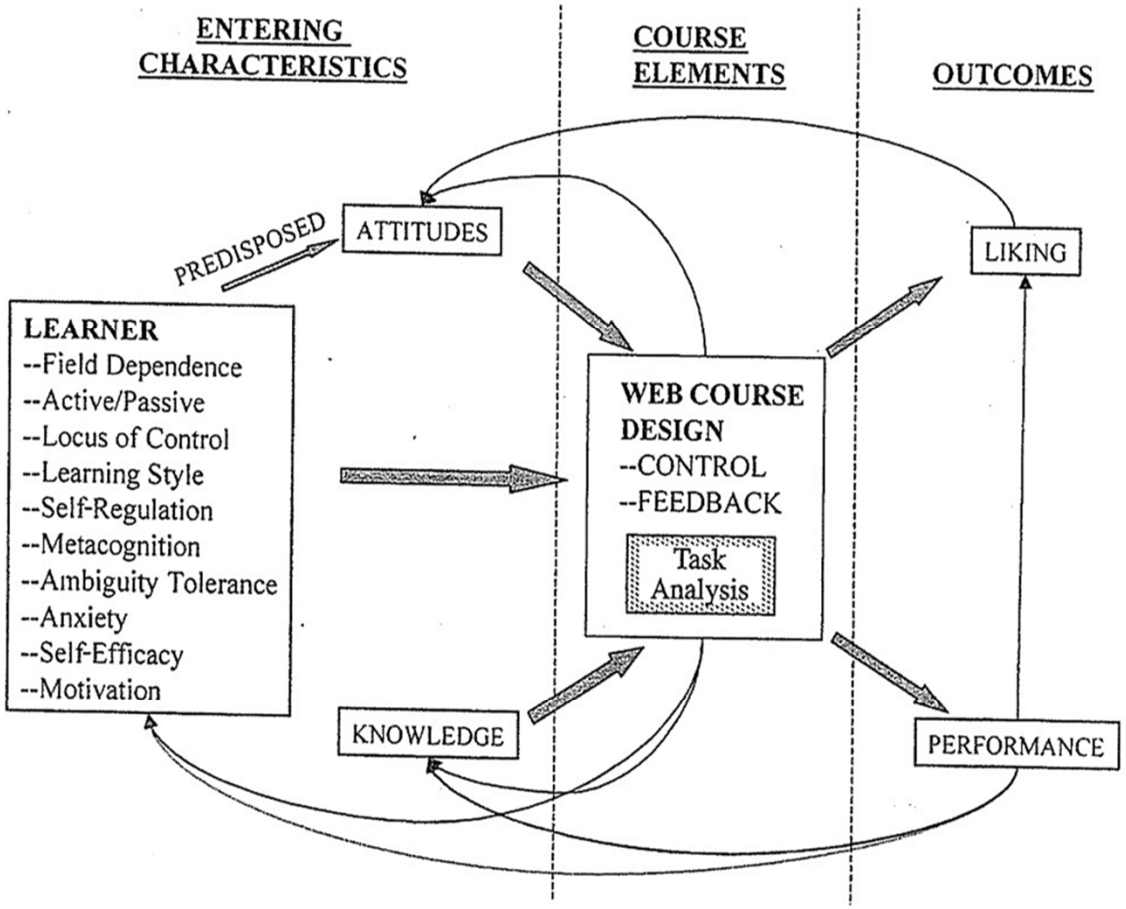

FIGURE 3.1.
Interactive model of learner characteristics and Web-based courses.

Personality Dimensions

- Ambiguity Tolerance
  - Ambiguity tolerance refers to an individual's willingness to accommodate or adapt to ambiguous situations or ideas.
  - Ambiguity situations develop from **novel circumstances** (no familiar cues), **complex situations** (many cues to be considered), contradictory **situations** (different elements or cues suggest different structures), **unstructured situations** (cues can't be interpreted).

Personality Dimensions

- Ambiguity Tolerance
  - Tolerant individuals accept new situation in which the rules or procedures are not known.
  - These individuals perform well when faced with a learning situation containing a great deal of novelty, complexity, contradiction, lack of structure, complex problem solving, and divergent learning approach.
  - Individual with low ambiguity tolerance rely on feedback and seek it out whenever possible.

Personality Dimensions

- Anxiety
  - It can be characterized as either a “trait” or “state” , representing individual emotional mood or condition.
  - There is a **curvilinear** relationship between anxiety and performance.
  - **Low-anxiety** students are more likely to benefit from instruction that encourages s**elf-regulation tasks**, **provides changes in routine**, **and is visually based**.
  - **Highly-anxiety** students prefer a situation with **structured routine** and **externally imposed goals**.

Personality Dimensions

- Field Dependence/Independence
  - Field-independent person is characterized as analytic, **self-referent**, and **impersonal** in orientation.
    - Field independent students have clear ideas of their own standards and values, and are logical, organized, and analytical when approaching intellectual activities.
  - Field-dependent person is characterized as global, socially sensitive, and interpersonal in orientation.
    - Field-dependent students are influenced by their physical and social backgrounds, and are less able to rely on their own judgments, organize their own learning activities, and maintain their own direction.

Personality Dimensions

- Field Dependence/Independence
  - Field-independent students are more self-directed learners, and performed better using discovery instructional methods, like distance education, that provide minimum guidance.
  - Field-dependent students performed better with maximum guidance.

Personality Dimensions

- Field Dependence/Independence
  - Field independent processors generally prefer to impose their own structure rather than have the structure imposed externally.
    - better hypertext processors (less structured).
    - more effective learners from hypermedia-based instruction.
  - Field-dependent learners prefer a slower pace, generate fewer combinations, recall fewer cues, and are more passive in a multimedia environment.

Personality Dimensions

- Active/passive leaners
  - Active learners perform especially outstanding of all others in a hypermedia environment (flexible not linear) →require more active search and construct, self-monitor
  - Passive learners seem to depend on prompts, feedback and explicit cues (**existent and organized**)

Personality Dimensions

- **Locus of Control (Attributions of Success and Failure)**
  - Students with internal control are more likely to demonstrate a deep approach to learning.
  - Perceived having control over their own learning, students are more likely to use **information processing** approaches: focus on the content as **whole**, try to see **connections** between the parts, and actively **organizations** of the information.
  - Perceived lack of control is most likely to lead to the view of learning as a memory task.

Personality Dimensions

- **Locus of Control (Attributions of Success and Failure)**
  - Internal controllers predicts better academic performance, because they believe that they can act on their own behalf, and have better study habits, whereas externals perform better under teacher control.
  - Students with an internal locus of control are better able to structure their navigation and take advantage of hypertext learning environments..

Personality Dimensions

- Self-efficacy
  - Self-efficacy, the belief in capability to execute actions toward designed goals.
  - Self-efficacy is one of the most powerful factors in predicting students' uses of learning strategies, and performance.
  - Students' beliefs in their ability to self-regulate their learning affected their perceived self-efficacy for academic achievement.

Learning style

- Depth of cognitive processing
  - Deep approach is typified by an interest in the academic task, an enjoyment in carrying it out, and a concentration on its relationship to previous knowledge.
  - Students with surface approach see the task a demand to be met, relies on memorization of the factual components, and concentrates on the discrete components of the task unrelated to other tasks.
  - Deep processors were able to consistently produce more effective study behaviors in a hypertext environment.

Learning style

- Instructional Modality
  - Paivio's dual coding theory, which addresses a person's preferred method of processing information (verbal Vs. imagery subsystems).
  - Although everyone codes information in both subsystems to some extent, individuals differ in their preferred representational style.
  - Combined use of audio and visual information results in more recall than visual-only or audio-only presentations.

Learning style

- Instructional Modality
  - Students with visual preference take advantages of imagery-intensive materials, such as pictures, graphs, and charts, and are proficient at video games.
  - Verbal learners succeed with materials that is presented in lectures, tapes, and nonillustrated texts.
  - Research shows that students who prefer a graphical learning setting benefit most from multimedia presentations, while students who prefer verbal presentations may actually be hindered by multimedia.

Executive Cognitive Processes

- Metacognition (thinking about thinking)
  - Research consistently shows a positive relationship between metacognition and academic performance.
  - Students who are aware of one or more metastratgeies for goal attainment are effective learners.
  - Research concludes that distance learners often fail to monitor their progress and comprehension of course material. That is, the lack of metacognition results in poor results.

Executive Cognitive Processes

- Self-regulation
  - Self-regulated learners are characterized as those students who rely on their internal resources to control their own learning process.
  - Self-regulated learners proactively seek out information, and are motivationally, behaviorally, and metacognitively active participants in their own learning.
  - Subprocessess of self-regulation consist of self-observation, self-judgment, and self-reaction.

Malleability of Individual Characteristics

- Personality characteristics
  - Research shows that some individual characteristics may be responsive to some forms of intervention.
  - A study demonstrates that nursing students become more field independent (more self-directed) during their participation in a distant education program.
  - Students in individualized computer-based learning become more internal because they can relate success to their own efforts.

Malleability of Individual Characteristics

- Approach to learning
  - A study reports the results of involving students in creating Web-based learning modules-an activity which forced them to become familiar with the nonlinear mechanisms of hypertext.
  - As a result of instructional format, students show a significant increase in their use of deep learning strategies as compared to their use prior to engaging in their activity.

Malleability of Individual Characteristics

- Executive cognitive processess
  - Self-regulated learning is perceived as a ‘developable aptitude”.
  - High level of self-regulatory automaticity is only possible once the student has become proficient at a task, and no longer needs conscious effort for self-direction.
  - A novice in a domain cannot be expected to display the same sorts of self-regulation that an expert would employ.

Malleability of Individual Characteristics

- Executive cognitive processess
  - The external regulation needs to be presented in such a way as to provide a scaffolding, structure for the student---the external one can be gradually removed as the student's own self-regulation develops.
  - Research shows a need to include mechanisms and feedback to improve students' self-regulation in a Web-based tutorial lesson.

Characteristics of Experience

- Knowledge (Declarative and Procedural)
  - The difference between novices and experts is readily evident in the manner in which they navigate through hypertext environments.
  - Students with greater domain knowledge (experts) show more purposeful navigation and allocate time more variably to different information nodes in hypertext environment.

Characteristics of Experience

- Knowledge (Declarative and Procedural)
  - Experts in a domain distinguish from novices by the nature of their mental models and how to use them to represent and manipulate knowledge.
  - One of the significant effects of the computerized presentation of hypertext material is the mapping of the experts' cognitive structures onto information to be presented.

Characteristics of Experience

- Knowledge (Declarative and Procedural)
  - If students are directed explicitly to attend to the experts' cognitive structure, the novice is able to acquire the structural knowledge.
  - Hypermedia's support of structural mapping would ideally lend itself to helping novices acquire an expert's representation of a subject domain.

Characteristics of Experience

- Attitude
  - Students with more positive attitude toward computer perform better than students with negative attitude.
  - A study in distance courses concludes that ability is a more critical factor in achievement for younger students, while attitudinal, motivational and personality factors may contribute more to achievement among older students.

Characteristics of Experience

- Attitude
  - Previous experience is the best predictor for attitude toward computers.
  - Most students' attitude improve with exposure to the use of enhanced media.
  - Research shows that even though multimedia does not uniformly lead to higher recall, positive attitudes resulted from the presentation regardless of the students' preferred presentation style.

Interactive model of learner characteristics and web-based courses

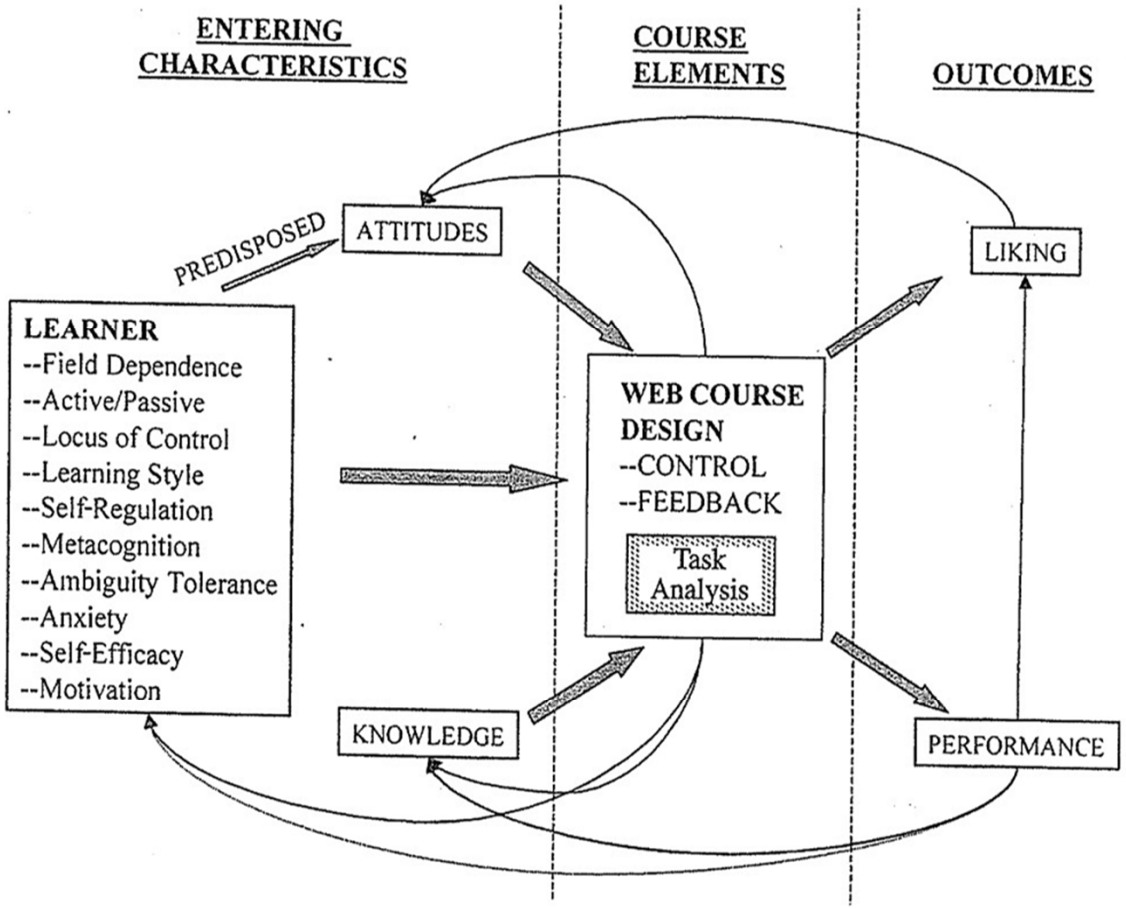

FIGURE 3.1.
Interactive model of learner characteristics and Web-based courses.

Salient Elements of Web-Based Course Design

- Control
  - One of the course design elements that varies the most is the amount of control the program offers the student, in terms of both navigational freedom and pacing of course -related activities.
  - The ability to effectively control the learning process results from an interaction of the student's independence, competence, and support (both human/emotional and resource/content).

Salient Elements of Web-Based Course Design

- Control
  - High levels of external control help students with low ability, but hinder those with high ability.
  - Students who combined self-regulation with deep processing benefit from limited external structure, while students who demonstrate surface processing and less-self-regulation require additional structure in a hypertext learning situation.

Salient Elements of Web-Based Course Design

- Control
  - For students who do not have the native ability benefit from the freedom of environment, appropriate external controls can be designed in the system itself.
  - A critical element in the transition from text to hypertext is the removal of a forced linear presentation of material.

Salient Elements of Web-Based Course Design

- Feedback
  - Availability of feedback while engaged in the task is critical to the development of self-monitoring skills.
  - Self-regulating learners are able to effectively generate internal feedback while engaged in an activity, and use that feedback to monitor their progress toward their goals.
  - For students who are not self-regulated, it is suggested to have mechanisms for generating external feedback built into their instructional modules.

Salient Elements of Web-Based Course Design

- Feedback
  - The external feedback protocols should stimulate students to develop the ability to self-monitor and become sensitive to intrinsic feedback, thus reducing their dependence on external support.
  - Research concludes that learning improves when feedback informs students about their monitoring of learning needs and guides them in how to achieve learning objectives.

Salient Elements of Web-Based Course Design

- Task characteristics
  - A task characteristics needs to clearly state the goals of the activity and identify the learning strategies most suitable for achieving those goals.
  - Course designers need to be **expert**s in the field, and build the supporting "**scaffolding**" for using those suitable strategies into the program.

Salient Elements of Web-Based Course Design

- Task characteristics
  - The task structure helps the novice accomplish the task, but it should be flexible enough to allow the expert to bypass the externally imposed structure and complete the task according to his/her preferred heuristic.
  - One of the basic conclusions is that any course is only as good as the material used to support it---with poor materials even good students cannot succeed.

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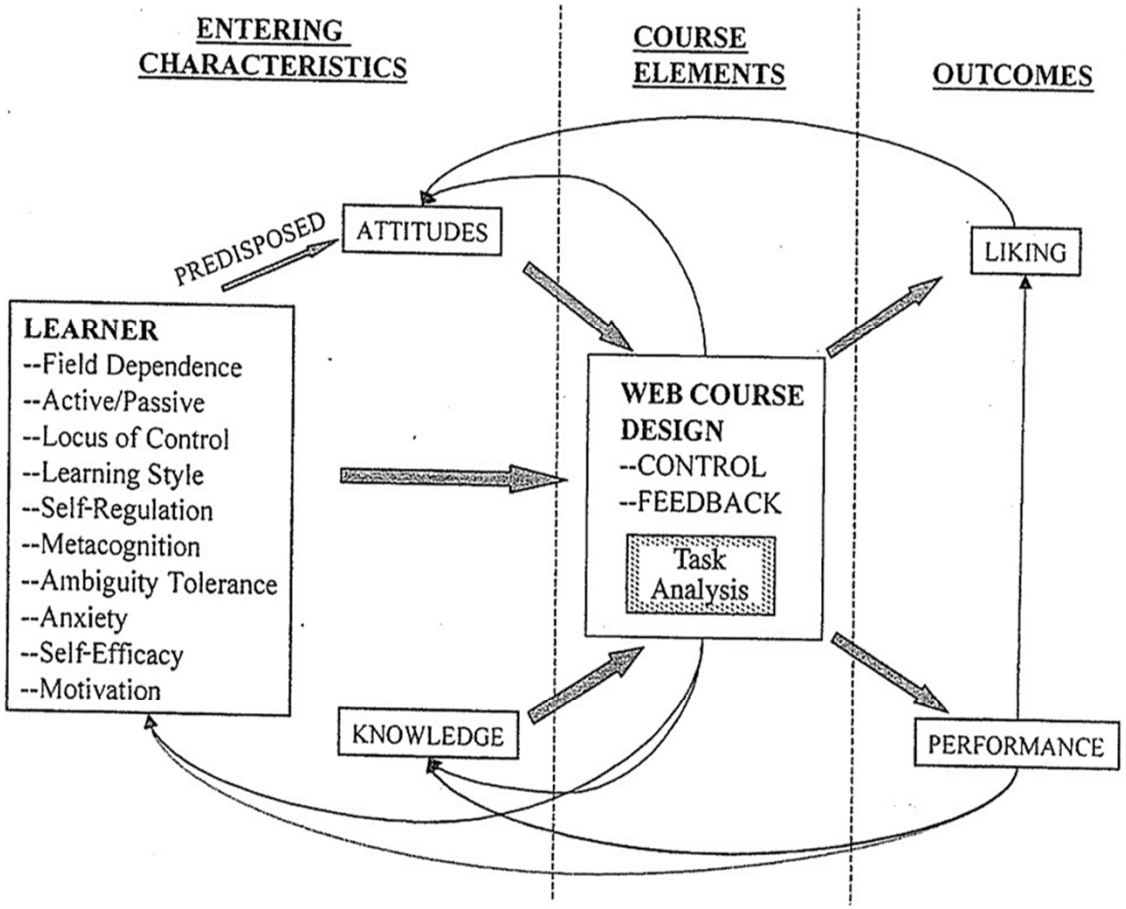

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